

SANAE EL-BYARI

Software Engineer · Recent Graduate

+212 681 768 607 · elbyarisanae@gmail.com · Rabat, Morocco · [LinkedIn](#) · [Portfolio](#)

PROFILE

Graduate of the École des Sciences de l'Information (ESI) in Rabat, specializing in Software Engineering, with training focused on software architecture design and modeling of complex information systems. Drawn to digital transformation and artificial intelligence challenges, where thinking through the solution takes precedence over pure technical execution.

PROFESSIONAL EXPERIENCE

Final-Year Project Intern – Migration of Payments & Collections to Microservices Feb. 2026 – Jul. 2026

Sinone Expert Univers – Rabat (on-site) · Grade: A

- Analyzed the financial processes of the Polysco monolithic system and designed the target microservices architecture.
- Developed payment microservices (receipts, checks, collections): Spring Boot, REST API, Feign REST, Apache Kafka.
- Secured APIs with Keycloak OAuth2/JWT, RBAC role management, transaction logging and auditing.
- Built the React.js frontend (Vite, TanStack Router, Tailwind CSS): payment and collection modules, automatic blocking of unpaid invoices.
- Set up a Jenkins CI/CD pipeline (SonarQube, Docker Build, Harbor) and Docker containerization.
- Result: 5 operational modules, defended with Grade A.

Stack: Spring Boot · Apache Kafka · Keycloak · React.js · TanStack · Tailwind CSS · Docker · Jenkins · SonarQube · PostgreSQL

Developer Intern – Citizen Safety Mobile App (SafeFes)

2025 · 2 months

SOFRETIM – Casablanca

- Developed a mobile app allowing citizens to anonymously report incidents (fires, violence, assaults, etc.) in real time.
- Integrated photo and video capture directly within the app, along with user-entered incident details.
- Implemented automatic geolocation capture and instant transmission to the relevant local authorities.
- Designed a secure back-office with role-based access control (RBAC) enabling authorities to view incidents on a map, review submitted media, and take action.
- Built Streamlit dashboards to monitor and analyze reported incidents.
- Developed a safe-route feature: detecting high-risk zones and suggesting a safer alternative route to the user's destination.
- Centralized and synchronized data via the Orion Context Broker (FIWARE) and MongoDB through a REST API.

Stack: React Native, Streamlit, FIWARE, MongoDB, REST API

Developer Intern – Digitalization of Performance Review Meetings

2024

OCP – Jorf Lasfar

- Developed a web application for automatic import of Excel files and data extraction using Pandas.
- Designed a dashboard displaying KPIs and dynamically generated performance charts.
- Automated one-click PowerPoint report generation (python-pptx).

Stack: Python: Flask, Pandas, python-pptx

Web Developer Intern

2025 · 3 months

MediaTower – Fès (remote)

- Developed the frontend of a Moroccan e-commerce platform showcasing products through videos, on the model of Jumia or Alibaba.
- Integrated interactive video players allowing buyers to view products directly from product pages.
- Built a dedicated seller dashboard: product, order, and sales management.
- Designed an interactive shopping cart and a smooth product search and navigation system.
- Set up real-time analytics dashboards displaying key KPIs (orders, revenue, seller performance).

Stack: React.js

ACADEMIC PROJECTS

Food Delivery Platform – Microservices Architecture

Module capstone project

- Designed and developed a complete food-delivery platform connecting customers, restaurant owners, and administrators.
- Java/Spring Boot backend in a microservices architecture, REST APIs secured with JWT, PostgreSQL database.
- Flutter/Dart mobile app: sign-up, restaurant browsing, cart, order confirmation, and real-time tracking.
- Angular back-office: admin dashboard, owner validation, product management, and platform-wide statistics.
- Docker/Kubernetes containerization, API testing with Postman, Git/GitHub versioning.

Stack: Spring Boot · Spring Security/JWT · PostgreSQL · Flutter/Dart · Angular · TypeScript · Docker · Kubernetes · Postman

Multi-Cloud Platform for AFCON 2025 – Cloud Architecture

Cloud Computing module project

- Designed and deployed a service-oriented architecture combining three cloud models (FaaS, PaaS, IaaS) to simulate a sports information platform (AFCON 2025).
- Developed serverless backends with AWS Lambda (Python), exposed via API Gateway: REST endpoints for teams, matches, news, and bookings.
- Hosted a ranking service on an Azure VM (IaaS) with Nginx, network configuration (SSH/HTTP), and JSON data deployment.
- Developed and deployed a React.js frontend on Firebase Hosting (PaaS), with state management, routing, and multi-cloud API consumption via a centralized service.
- Set up continuous integration and inter-service communication via HTTP/JSON requests.

Stack: AWS Lambda · API Gateway · Azure VM · Nginx · React.js · Firebase · Python · JavaScript

E-commerce Performance Analysis and the Impact of Product Returns

Business Intelligence project

- Designed a decision-support solution to analyze the profitability of an e-commerce platform and the impact of product returns on financial losses and customer satisfaction.
- Modeled a data warehouse using a star schema (fact and dimension tables) from a synthetic dataset (Faker).
- Built ETL workflows with KNIME for data extraction, cleaning, transformation, and loading.
- Created interactive dashboards and analytical reports with Metabase and Power BI, covering key KPIs (net revenue, return rate, financial losses, customer churn).
- Performed multidimensional performance analysis by product category, age group, and payment method.

Stack: Python (Faker) · KNIME · SQL · Metabase · Power BI · Star schema modeling · ETL

Embedded Driver Drowsiness Detection System

Embedded Systems project (STM32)

- Designed and developed a real-time system on STM32 (NUCLEO-C031C6) to detect driver drowsiness via eye closure and head movement.
- Integrated and programmed sensors: MPU6050 (IMU) for head tilt and a photoresistor (LDR) simulating eye-closure detection (replacing the TCRT5000 for Wokwi simulation).
- Implemented a 4-state detection logic (Normal → Attention → Fatigue → Alert) with configurable thresholds.
- Drove actuators: 16x2 LCD display for status messages, LEDs (Green, Yellow, Red, Blue) for visual alerts, and a buzzer for audio alerts.
- Optimized the embedded C code to reduce memory footprint and ensure real-time performance on the microcontroller.
- Fully simulated and validated the wiring on Wokwi before implementation.

Stack: STM32 (NUCLEO-C031C6) · C · MPU6050 · LDR · Wokwi · 16x2 LCD · Buzzer

Legal Consultation Mobile Application

- Mobile app for legal consultation with a Spring Boot backend and React Native frontend.

Agricultural Management Dashboard – Power BI

- Interactive Power BI dashboard for managing and visualizing data from an educational farm (livestock, sales, production).

Event Management CRUD Application – JavaFX & JDBC

- Desktop CRUD application for managing customers, menus, and events. JavaFX interface, PostgreSQL database via JDBC.

Hospital Stay Management Application – Java

- Desktop Java application (JDBC, JavaFX/Swing) for managing hospital stays and patient records, PostgreSQL.

Intelligent Knowledge Base with AI

- Guidance system for high-school students featuring a conversational chatbot powered by a dedicated AI platform.

Wordle-style Game in C with SDL

- Wordle-style game in C using the SDL graphics library: interaction handling, dynamic display, optimized game logic.

TECHNICAL SKILLS

Languages & Frameworks: Java (JavaFX, JDBC, Maven), JavaScript, TypeScript, React.js, React Native, Dart (Flutter), Python (Flask, Pandas, Streamlit), C (Embedded), PHP, HTML/CSS, SQL

Backend & APIs: Spring Boot, Flask, AWS Lambda (FaaS), API Gateway, REST APIs, Apache Kafka, Feign REST

Databases: PostgreSQL, MariaDB, MongoDB

Security: Keycloak, OAuth2/JWT, RBAC

DevOps & Tools: Git/GitHub, Docker, Kubernetes, Jenkins, SonarQube, Harbor, Maven

Cloud Computing (Multi-Cloud): AWS (Lambda, API Gateway), Microsoft Azure (IaaS – VM, Networking), Google Cloud (Firebase Hosting), FIWARE (Orion Context Broker)

Data & BI (Business Intelligence): Power BI, Metabase, ETL/ELT (KNIME), Star schema modeling, OLAP, Data visualization, Excel automation, Reporting

Embedded Systems & IoT: STM32 (NUCLEO-C031C6), Sensors (MPU6050, LDR), Wokwi (Simulation), Actuators (16x2 LCD, Buzzer), Real-time programming (C)

Mobile & UI/UX: React Native, Flutter, Mobile UI/UX Design

SOFT SKILLS

Automation & Optimization: Workflow automation, reporting pipelines, real-time dashboards

Project Management: Agile collaboration, cross-functional teamwork, planning and coordination

Problem Solving: Algorithm design, debugging, code and process optimization

Communication: Technical documentation, presenting analyses to stakeholders

EDUCATION

Engineering Degree – Software Engineering 2023 – 2026

École des Sciences de l'Information (ESI) – Rabat · Track: Information Systems Development & Digital Transformation

Preparatory Classes for Grandes Écoles (CPGE) 2021 – 2023

Mathematics & Physics · Admitted after two years of CPGE

LANGUAGES

Arabic — Native · French — Fluent · English — Intermediate